

Logarithmic differentiation - 0.3.9

Example: Compute $\frac{d}{dx} \ln\left(\frac{x^2 \sin(x)}{2x+1}\right)$.

$$\begin{aligned} \frac{d}{dx} \ln\left(\frac{x^2 \sin(x)}{2x+1}\right) &= \frac{2x+1}{x^2 \sin(x)} \cdot \frac{d}{dx} \left(\frac{x^2 \sin(x)}{2x+1} \right) \\ &= \text{quotient rule} \quad \text{"} \end{aligned}$$

Instead, use log rules:

$$\frac{d}{dx} \ln\left(\frac{x^2 \sin(x)}{2x+1}\right) = \frac{d}{dx} \left(\ln(x^2 \sin(x)) - \ln(2x+1) \right)$$

$$= \frac{d}{dx} \left(2 \ln(x) + \ln(\sin(x)) - \ln(2x+1) \right)$$

$$= \frac{2}{x} + \frac{1}{\sin(x)} \cos(x) - \frac{1}{2x+1} \cdot 2 = \frac{2}{x} + \cot(x) - \frac{2}{2x+1}$$

Example. Compute $\frac{d}{dx} x^x$.

→ Can't use the power rule $\frac{d}{dx} x^n = nx^{n-1}$
since the exponent is not constant

→ Can't use the exponential rule $\frac{d}{dx} a^x = a^x \ln(a)$
since the base is not constant

How to attack this? Take a logarithm.

$$y = x^x \Rightarrow \ln(y) = \ln(x^x) = x \ln(x)$$

$$\ln(y) = x \ln(x) \Rightarrow \frac{1}{y} y' = \ln(x) + x \cdot \frac{1}{x} = \ln(x) + 1$$

$$\Rightarrow y' = y(\ln(x) + 1) = x^x (\ln(x) + 1).$$

Example: Find $\frac{d}{dx} \left(\frac{\tan(x)}{x^2+1} \right)^{\sin(x)}$.

$$y = \left(\frac{\tan(x)}{x^2+1} \right)^{\sin(x)} \Rightarrow \ln(y) = \sin(x) \ln \left(\frac{\tan(x)}{x^2+1} \right)$$

$$\ln(y) = \sin(x) \left(\ln(\tan(x)) - \ln(x^2+1) \right)$$

$$\frac{1}{y} y' = \cos(x) \left(\ln(\tan(x)) - \ln(x^2+1) \right) + \sin(x) \left(\frac{1}{\tan(x)} \sec^2(x) - \frac{1}{x^2+1} \cdot 2x \right)$$

$$y' = \left(\frac{\tan(x)}{x^2+1} \right)^{\sin(x)} \left(\cos(x) \left(\ln(\tan(x)) - \ln(x^2+1) \right) + \sin(x) \left(\frac{1}{\tan(x)} \sec^2(x) - \frac{1}{x^2+1} \cdot 2x \right) \right)$$